

SURFACE PROXY · SPATIAL ANALYSIS · OPEN WORKFLOW

Surface titanium and lunar crustal magnetism

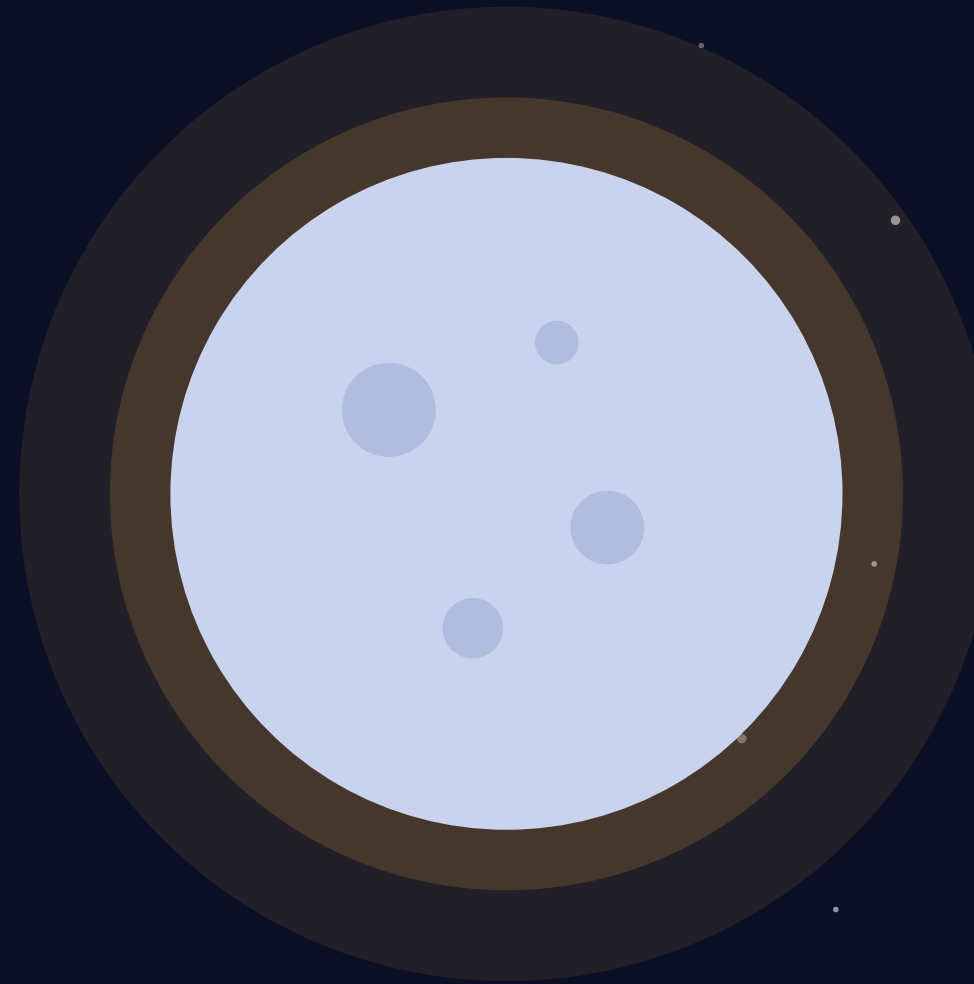
- Global-map analysis over the declared joint-valid footprint (LROC $\pm 70^\circ$ and Imbrian-age restriction). **It does not directly test** the dated temporal mechanism of Nichols et al. (2026).



EDUCATIONAL & AI-ASSISTED

This presentation and the accompanying repository are a personal project created for learning and hands-on experience.

Artificial Intelligence (Large Language Models) was used extensively throughout its development to help write the code, understand the geophysical concepts, and rigorously catch statistical and logical errors.



THE PUZZLE

A magnetized crust without a present-day global dynamo

- **No global field today**

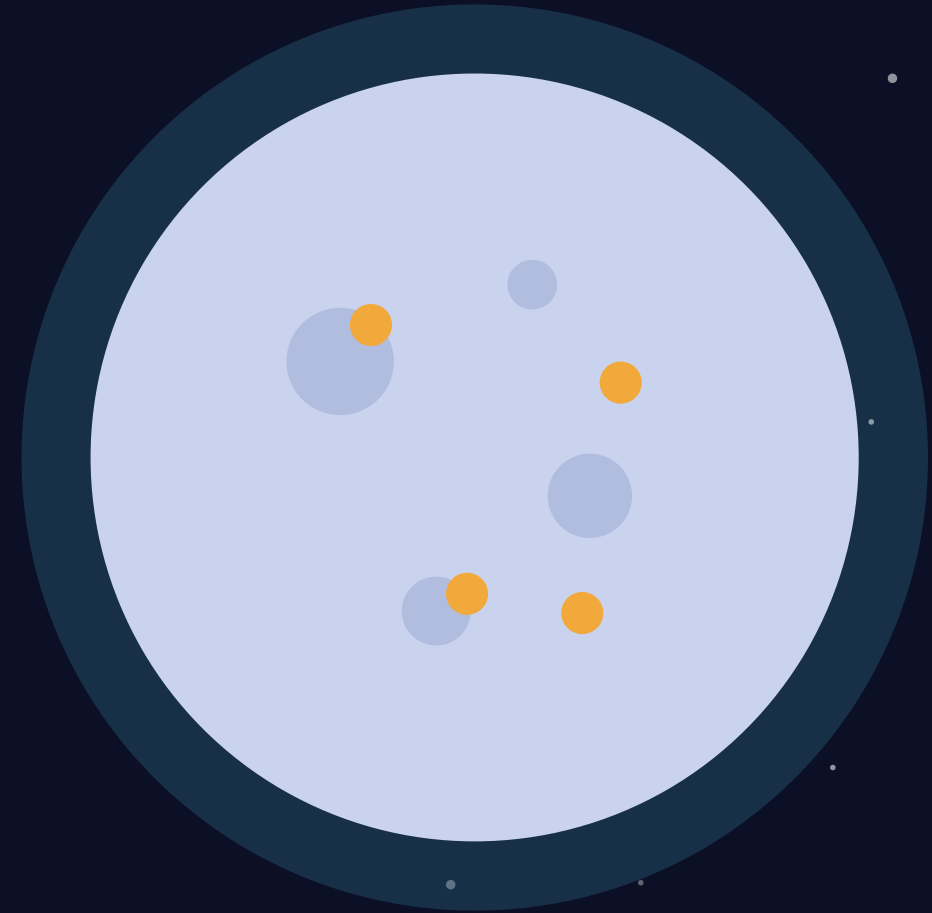
Unlike Earth, the Moon has no active global core dynamo today.

- **Yet the crust is magnetized**

Orbiting spacecraft measure strong, patchy magnetic anomalies locked into ancient rock.

- **Timing is the missing variable**

Dated samples constrain ancient field evolution. Present-day surface maps test spatial co-location, not the temporal dynamo mechanism.



THE SCOPE

One temporal mechanism motivates one spatial proxy test

● Motivation · dated samples

Nichols, Wade & Stephenson (2026), Nature Geoscience

Favored mechanism: radiogenic melting of ilmenite-bearing cumulates at the core-mantle boundary after overturn, raising core heat flux (not simply continued sinking of Ti-rich material).

Temporal mechanism

● Benchmark · impact antipodes

Hood & Artemieva (2008), Icarus

Fold-matched H2 PR-AUC .271820 vs null mean .155877, SD .103869 and 95th .376621 ($p=.172414$ (not recovered)).

86 fold-matched shifts · 5/5 folds evaluable

Does surface TiO_2 add spatial information for crustal magnetism after mapped geologic controls?

1

Repository plan

Ad hoc declared analysis plan; timing cannot be independently verified (repo history begins after results). Binding scores use default-config XGBoost; nested tuning is diagnostic only.

2

Power-gated

Positive and negative outcomes are explicit; negative claims require adequate detection power.

3

Scope-limited

The result concerns spatial co-location—not Nichols et al.'s temporal dynamo mechanism.

THE EVIDENCE

Five principal products (22 pinned source files), plus a mapped mare-proxy mask

- **Magnetic field (what we predict)**

Tsunakawa et al. 2015 surface SVM via Wieczorek T2015_449 (surface-evaluated $|B|$; primary threshold 10 nT)

- **Surface titanium — TiO_2 (proxy)**

LROC WAC UV/Vis; values <2 wt% are non-quantitative (product floor); primary uses ≥ 2 wt% only

- **Bouguer gravity**

GRAIL GRGM1200A — subsurface density structure

- **Crustal thickness**

GRAIL crustal-thickness archive

- **Geology + mare proxy**

USGS units define age and mapped mare-proxy scope

~2 GB

of institutional data

23

pinned source files

SHA-256

every byte verified

1° grid

≈ 30 km at the equator

THE METHOD

Four stages separate description from inference

1

Align

Align five maps and the derived mapped mare-proxy mask on one equal-area lunar grid.

2

Describe

Fit continuous amplitude with controls, then add Ti features: raw + 25/50/100 km.

3

Restrict

Repeat controls vs. controls+raw-TiO₂ only inside the mapped USGS mare-proxy mask.

4

Calibrate

Match each statistic to its own spatial null, then inject H₂ signals to measure power.

Why blocking still falls short: 30° (~910 km) and 60° (~1819 km) holdouts remain below the ~3752 km dependence range. Both nulls used the first 100 eligible 5/5-fold unique nonidentity shifts from 125 candidates; 12 four-fold candidates were rejected. Primary prevalence $\approx$.0384 (10 nT, quantitative TiO₂).

Reproducible evidence still has design limits

● Repository self-attestation

Author-declared plan only — not an independently timestamped preregistration. v1.0.0 is superseded.

● Logical-fallacy audit

Inference traps are documented and corrected in the open—including the matched H2 null.

● Fail-closed provenance

Hashes, real/synthetic guards and mare-mask alignment are enforced in code.

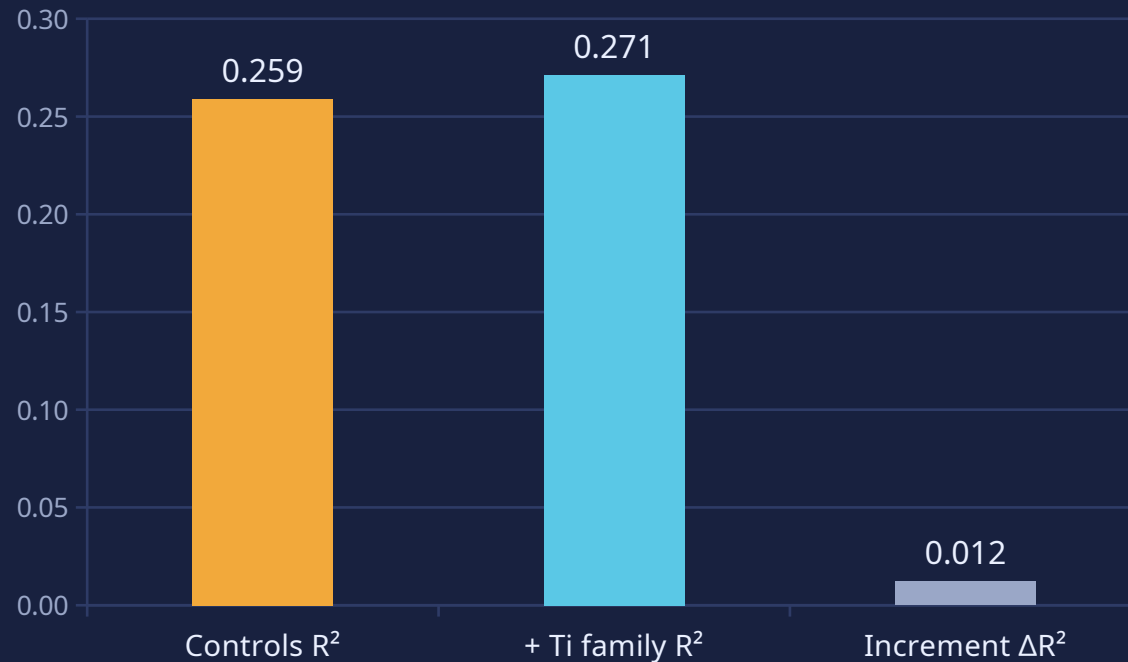
● Automated tests

Guards cover provenance, terrain validity, power gating and report claims.

THE RESULT

The full Ti family adds only a small descriptive increment

Held-out continuous R^2 (descriptive)



ΔR^2 **-.0146**

controls .3649 → .3503 with full Ti family; descriptive only

Δ PR-AUC
+.042444

remove raw/buffered Ti + Ti×gravity interactions; Wilcoxon p unavailable

$n_{\text{eff}} \approx 1$

30° 910 km holdout < 3752 km dependence range

STATUS: INCONCLUSIVE_LOW_POWER

• **failure-to-detect, not disproof**

This spatial proxy result neither confirms nor refutes Nichols et al.'s temporal mechanism.

TERRAIN VALIDITY

Mapped mare-proxy TiO_2 remains descriptive and unstable

Mapped mare-proxy binary

$\Delta\text{PR-AUC} +.055306$; $+\text{TiO}_2$.476561 versus controls .421255 (mare descriptive only).

n=3,928 · 165 positives



Mapped mare-proxy continuous

Primary continuous R^2 .3649 $\rightarrow$.3503; raw-Ti increment $+.014559 \pm .012575$. Common-fold gain vs full raw-Ti: $+.001445$.

proxy range 3289 km · $n_{\text{eff}}=1$

Interpretation: Report binary and continuous estimands separately. Registered binary mare criterion did not pass; continuous increments stay descriptive. Footprint still supplies ~one dependence-scale unit.

Three diagnostics set the inference ceiling

- **Dependence exceeds blocks**

$n_{\text{eff}} \approx 1$. Even 60° blocks (~1819 km) remain below the ~3752 km dependence range.

- **Matched H2 benchmark recovered**

Full PR-AUC .394936 vs fold-matched null mean .231862, SD .103869 and 95th .490983 ($p=.137931$); H2 match $p=.172414$ (not recovered).

- **Power floor is not adequate power**

H2 recovery at $\beta=0/.5/1/1.5/2/3-4$: 3/30, 18/30, 27/30, 28/30, 29/30, 30/30. Point MDE $\beta=1$; Wilson-lower MDE $\beta=2$.

“

INCONCLUSIVE
failure-to-detect, not
disproof
LOW POWER

$\beta=1$ implies risk +.0557 and $OR \approx 607$; no justified target effect.

The next test must be time-resolved.



- Treat this as a surface-proxy spatial study—not a temporal dynamo test.
- Use dated samples or time-resolved proxies to test the Nichols mechanism directly.
- Expand calibrated TiO_2 coverage and use dependence-scale blocks before any negative claim.